

Report
Of
Python Full Stack Development With Project

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student:2410031425

Name of Student: KRISH GUPTA

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, Krish Gupta do hereby declare that the internship report titled “**Python Full Stack Development With Project**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Krish Gupta
2410031425

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature
Krish Gupta
2410031425

INTERNSHIP COMPLETION CERTIFICATE



PRESENTED TO

Krish Gupta

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

Python Full Stack Development With Project

During the internship, the learner demonstrated proficiency in:

- Introduction to Full Stack Development & Project Setup & Python Fundamentals for Backend Development
- Introduction to FastAPI for Backend APIs & Data Validation and Modeling with Pydantic
- Introduction to React and Frontend Fundamentals & React Hooks and State Management
- Building Reusable React Components and Styling & Integrating FastAPI Backend with React Frontend
- Database Integration with FastAPI (SQLAlchemy) & Advanced FastAPI Features and Best Practices
- Advanced React Patterns and State Management (Redux/Zustand) & Frontend Routing and Navigation
- Building Forms and Handling User Input in React & Testing Your Full Stack Application & Deployment Strategies for Full Stack Applications
- API Design Principles and Best Practices & Frontend Best Practices and Performance Optimization & Capstone Project: E-commerce Platform with User Authentication and Product Catalog
- **Project: Markdown to HTML Converter**

This certification recognizes the successful completion of the internship requirements and project work.



Issued on: Aug 30, 2026
Certificate ID: 2026-9573F1FE58

Milu Swain
General Manager
(Learning & Development)
EduSkills

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4. PROJECT DESCRIPTION

4.1 Introduction

Industry-oriented learning through internships plays a crucial role in bridging the gap between academic knowledge and real-world applications. This 8-week online virtual internship provided hands-on exposure to Python Full Stack Development and its application in modern software solutions. The capstone project undertaken during this tenure was the development of a Markdown to HTML Converter, providing deep insights into text parsing and backend logic.

4.2 Organization Profile

Organization Name: EduSkills Academy. Domain: EdTech and Skill Development. Services: Virtual Internships supported by AICTE. Vision: To empower students with industry-relevant skills.

4.3 Problem Statement

The Markdown to HTML Converter is a software utility designed to automate the translation of plain text formatted with Markdown syntax into equivalent HTML code. This project tackles the common need for a simple yet effective way to render Markdown content for web display. It systematically addresses key Markdown features, including the conversion of various header levels, text emphasis, inline code segments, and hyperlinks.

4.4 Project Objectives

The primary objective is to design and implement a robust solution for converting Markdown syntax to HTML utilizing modern tools, algorithms, and regular expressions in Python. Another key goal is to optimize performance, accuracy, and code quality using a sequential processing pipeline.

4.5 Scope of the Project

The current scope encompasses the parsing and transformation of core Markdown elements including headers, bold and italic emphasis, inline code snippets, and hyperlinks. The logic ensures that the sequential pipeline processes a given Markdown string and outputs valid, semantic HTML.

4.6 Technologies and Tools Used

The programming language used is Python along with core libraries like `re` for regular expression operations, `sys`, and `json`. Core concepts applied include string manipulation, prefix identification, regular expressions covering pattern matching and capture groups, and pipeline architecture.

4.7 System Architecture

The system is designed as a sequential processing pipeline. The input Markdown text is read line by line. For each line, the system first checks for header prefixes. Subsequently, it passes the text through a series of regular expression substitutions in a specific order starting with links, followed by bold text, italics, and inline code. Finally, the processed HTML string is formatted into a JSON response.

4.8 Methodology

The methodology involves identifying lines starting with specific hash symbols and transforming them into standard HTML heading tags. Text emphasis is handled using regular expressions for bold and italic text. Inline code is parsed by matching content within backticks and converting it to code tags. Links are converted utilizing capture groups to extract text and URLs to construct anchor tags. Finally, all steps are integrated into a full pipeline to ensure transformations do not interfere with one another.

4.9 Expected Outcomes

The expected outcome is a fully functional script that can accept a standard Markdown document and output well-formed HTML. This tool serves as a foundational component for content creation platforms, blogs, and documentation sites.

4.10 Certificates and Communication



Ref No: C17Y2026M081508389

Internship Offer Letter

Student Name : Krish Gupta
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Python Full Stack Development + Project
Internship Duration : June 2026 - August 2026
AICTE Student ID : 218161993126

We are pleased to inform you that you have been selected for the 8-Week AICTE – EduSkills Virtual Internship Program.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a week-wise structured learning plan.
- You will be required to complete weekly assessments to track your progress and understanding.
- Certain modules will require submission of project documentation and assignments.
- In the final week, you must appear for a Final Assessment Test.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a Virtual Internship Certificate.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Introduction to Full Stack & Python Fundamentals	Introduction to full-stack development concepts and environment setup. Learn Python fundamentals including data types, functions, OOP concepts, and backend basics.
Week 2	FastAPI & Data Validation (Pydantic)	Build backend APIs using FastAPI framework. Learn request handling, routing, dependency injection, and data validation using Pydantic.
Week 3	React Fundamentals & Hooks	Learn React basics including components, JSX, props, state management, useState, useEffect, and building dynamic user interfaces.
Week 4	Reusable Components & Frontend-Backend Integration	Create reusable React components and integrate frontend with FastAPI backend using API calls and dynamic data rendering.
Week 5	Database Integration with SQLAlchemy	Integrate databases using SQLAlchemy ORM, perform CRUD operations, manage models, and connect FastAPI with relational databases.
Week 6	Advanced React & State Management	Learn advanced React patterns, routing with React Router, and state management using Redux or Zustand for scalable applications.
Week 7	Testing & Deployment	Implement backend and frontend testing strategies. Learn deployment techniques using Docker and cloud platforms.
Week 8	API design & Front end optimisation Capstone Project – Full Stack Application Development & Capstone Project	Begin building a complete full-stack application (E-commerce platform) integrating FastAPI backend, database (SQLAlchemy), authentication system, and React frontend.

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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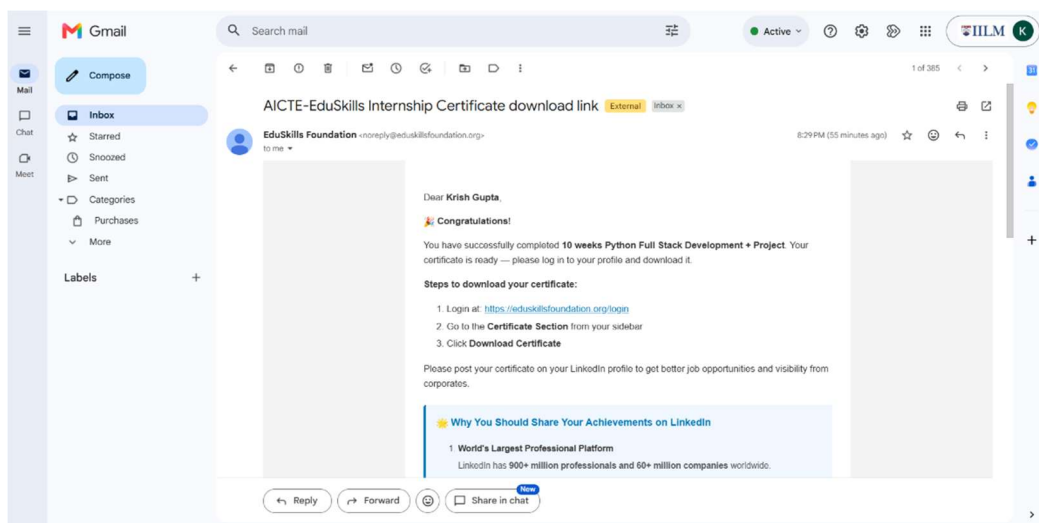
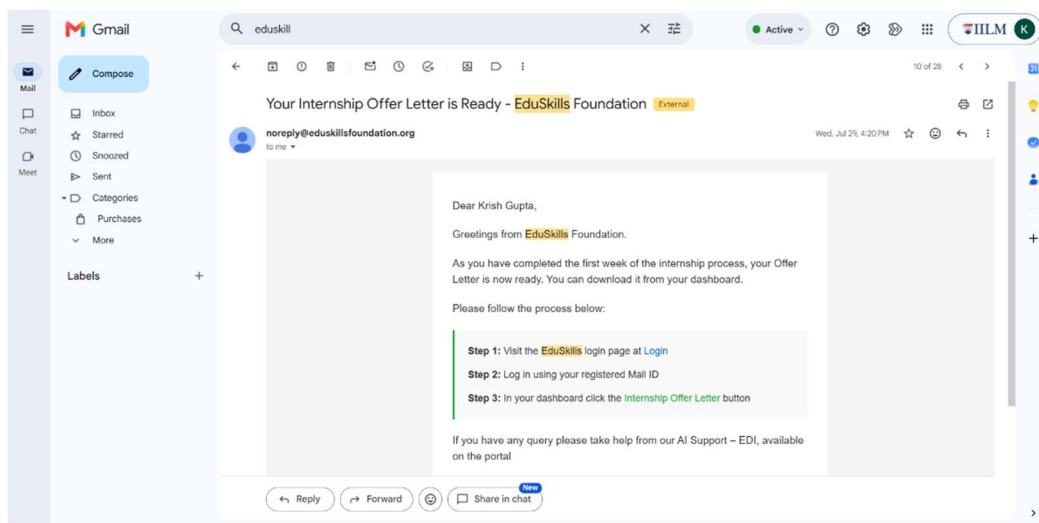
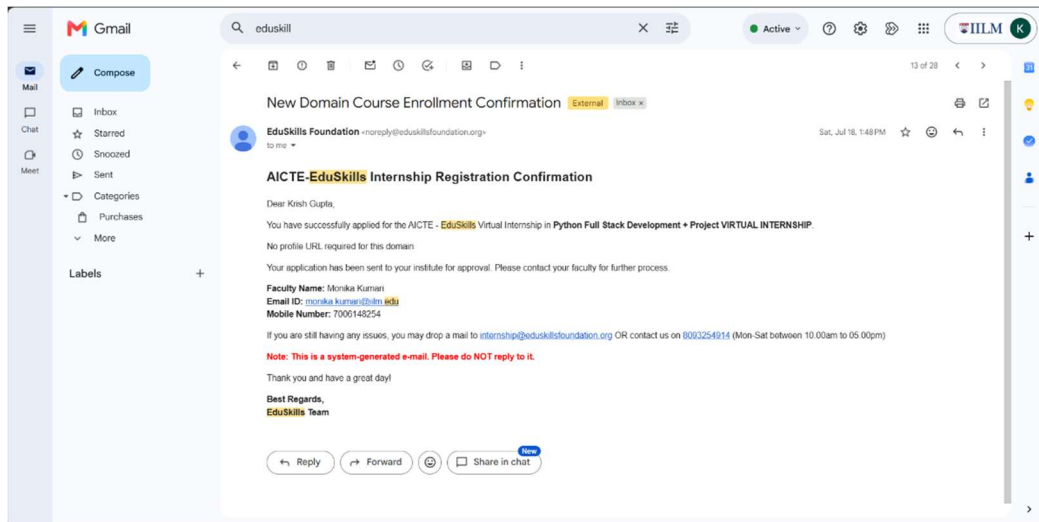


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5.BIBLIOGRAPHY/REFERENCES

- Python Software Foundation, Python 3 Documentation, Available at docs.python.org/3/
- Python Software Foundation, Regular Expression HOWTO, Available at docs.python.org/3/howto/regex.html
- Markdown Guide, Basic Syntax, Available at markdownguide.org/basic-syntax/
- EduSkills Academy LMS Learning Modules and Capstone Project Guidelines.